

# James Makhoul

Portfolio | LinkedIn | GitHub

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## MECHATRONICS ENGINEERING STUDENT

Fourth-year Mechatronics Engineering student with a perfect GPA and Dean's List recognition. Strong interest in control systems, robotics, and brain-computer interfaces (BCI), with hands-on experience in research, embedded systems, and software development. Actively engaged in interdisciplinary projects bridging mechanical, electrical, and computational engineering.

## EDUCATION

### Lebanese American University

Bachelor in Mechatronics Engineering

- Dean's Distinguished List
- GPA: 4.0/4.0

Byblos, Lebanon

Aug 2023 – Dec 2027

## EXPERIENCE

### Robotics Intern

inmind.ai

- Developed and executed SLAM simulation pipelines in Dockerized ROS 2 environments.
- Built and debugged multi-node ROS 2 packages integrating driver, filter, and logger nodes for real-time processing.

July 2026 – Aug 2026

Mkallès, Lebanon

### Engineering Intern

Mitsubishi

- Drafted 2D AutoCAD engineering drawings, including proposal layouts, shop drawings, and installation diagrams.
- Collaborated with the sales engineering team on cost estimation and drafting technical project commercial offers.
- Performed Quality Control (QC) field inspections on installed elevator systems, auditing mechanical integrity and logging snag reports.
- Designed a detailed 3D elevator and counterweight assembly in SOLIDWORKS, configuring mechanical mates to simulate cable motion.
- Constructed a functional 2-floor prototype elevator controlled by an Arduino Mega with IR sensors, limit switches, and motor actuators.

Jun 2026 – Aug 2026

Dbayeh, Lebanon

### Undergraduate Research

Lebanese American University

- Conducting research in Brain-Computer Interfaces (BCI), focusing on multi-channel EEG signal preprocessing.

Dec 2025 – Present

Byblos, Lebanon

## PROJECTS

### T.R.A.C.E. (Writing Robotic Arm)

Arduino, MATLAB/Simulink

- Designed and 3D-printed a 3-DOF robotic arm.
- Implemented a Task-Space Impedance Controller for precise trajectory tracking.
- Simulated on MATLAB using Simscape.

### Nine Lives (Game Development Project)

Flutter (Dart), Flame Game Engine, Firebase

- Developed a multiplayer and singleplayer pixel-art platformer using Flutter and the Flame engine.
- Implemented Firebase real-time database for real-time multiplayer gameplay.
- Hosted on <https://nine-lives-game.web.app/>

### Inverted Pendulum Control and Swingup

MATLAB/Simulink

- Designed and implemented swing-up and stabilization controllers by applying PID, fuzzy logic, LQR, and energy-based control strategies.

### Explorer Bot

Arduino, myRIO

- Developed an award-winning glove-controlled robotic system featuring a mechanical grabber, camera, and multiple sensors.

### Roommate Finder

Node.js, Express, JavaScript, CSS, HTML

- Designed and developed a web platform to match users with compatible roommates based on lifestyle, location, and budget.

## TECHNICAL SKILLS & SOFT SKILLS

- Engineering Tools:** MATLAB/Simulink, SOLIDWORKS, AutoCAD, ROS 2, CMake, Docker
- Programming and Development:** C++, Python, Java, Flutter (Dart), JavaScript, HTML, CSS, PostgreSQL, Firebase
- Machine Learning:** Deep Learning, Computer Vision, Neural Networks
- Soft Skills:** Problem Solving, Critical Thinking, Adaptability, Time Management, Teamwork

## CERTIFICATIONS & AWARDS

- Certified SolidWorks Associate (CSWA) by Dassault Systèmes
- LAU Engineering Projects Day Winner (Explorer Bot)
- 1st Place in the Lebanese National Mathematics Olympiad by CRDP

## VOLUNTEERING

- LAU Robotics Summer Camp 2025: Mentored high school students during hands-on robotics workshops.

## LANGUAGES

- English:** C2 | **French:** C1 | **Arabic:** C1